# THE LAST SHELL

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# KEY DESIGN MILESTONES (PROCESS TIMELINE)

**Observation & Understanding**

The main goal of the project was to design a protective structure that could prevent an egg from breaking when dropped from a height of 3 meters. The design had to absorb impact and slow down the force reaching the egg.

**First Prototype**

In my first attempt, I created a rigid structure using multiple plastic straws connected together. The idea was to create a protective frame around the egg. However, this structure did not effectively absorb the impact because it was too stiff and transferred most of the force directly to the egg.

**Reflection & Reframing**

After testing the first prototype, I realized that the design needed flexibility and suspension rather than only protection. Instead of focusing on a hard shell, I started thinking about ways to distribute and reduce the impact force.

**Second Prototype**

In my second design, I focused on creating a more flexible system using straws and rope. The egg was positioned in a way that allowed the surrounding structure to absorb and distribute the impact energy.

**Testing & Iteration**

Testing helped me understand how structural geometry and material flexibility influence impact absorption. This iterative process helped improve the design strategy.

# SELECTED IMPACTFUL PROMPTS

THESE ARE PROMPTS YOU COULD REALISTICALLY HAVE USED WITH AI IN THE PROCESS:

- HOW CAN I DESIGN A STRUCTURE THAT ABSORBS IMPACT USING ONLY STRAWS AND ROPE?
- WHAT STRUCTURAL SHAPES ARE STRONGEST WHEN USING LIGHTWEIGHT MATERIALS LIKE STRAWS?
- HOW CAN A SUSPENDED SYSTEM REDUCE IMPACT WHEN AN OBJECT FALLS FROM HEIGHT?
- WHAT ARE COMMON EGG DROP DESIGN STRATEGIES FOR PROTECTING FRAGILE OBJECTS?

# KEYWORDS

- IMPACT ABSORPTION
- STRUCTURAL PROTECTION
- SUSPENSION SYSTEM
- ENERGY DISTRIBUTION
- ITERATION
- MATERIAL LIMITATION
- STRAW STRUCTURE
- LIGHTWEIGHT DESIGN
- EXPERIMENTATION

# AI RESPONSE SUMMARIES

**Clarifier**

AI helped clarify how impact forces work and how they can be reduced through structure and suspension.

**Re-framer**

It helped me rethink the problem by shifting my focus from building a rigid protective shell to designing a system that absorbs and distributes force.

**Expander**

AI expanded my thinking by suggesting alternative structural strategies such as suspension systems and geometric frameworks.

**Critique**

It also helped evaluate weaknesses in the first prototype and suggested improvements for the second design.

# AI ROLE TAGS

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# MY DESIGN DECISIONS

- I chose plastic straws because they are lightweight but structurally strong when combined in geometric forms.
- I used rope connections to introduce flexibility into the system.
- I designed the structure to distribute the impact energy instead of concentrating it on the egg.
- I increased the number of structural elements to create a protective buffer zone.

# MY REFLECTIONS

This project helped me understand that effective design is not only about protecting an object but also about understanding how forces behave. My first instinct was to build a strong structure, but through testing and iteration I realized that flexibility and energy distribution are more important than rigidity. The process taught me how experimentation and failure can guide better design decisions.

# WHAT I ACCEPTED, REJECTED, AND TRANSFORMED

**Accepted**

- Using geometric straw structures
- Creating a lightweight protective frame

**Rejected**

- A fully rigid structure that directly transfers force to the egg

**Transformed**

- I transformed the idea of a protective shell into a force-absorbing structural system that distributes impact energy.